

Toward Turing’s Child Machine: Active Inference with LLMs in the Era of Experience

A live closed-loop demonstration of LLM-bridged Active Inference on ARC-AGI-3

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Implementation and writing assisted by Claude (Anthropic, Opus 4.7). All experimental design, claims, and conclusions are the author’s.

Abstract

In 1950, Alan Turing proposed building intelligence not by simulating an adult mind but by constructing a *child machine* that learns from its own experience. Three-quarters of a century later, Silver and Sutton’s *Era of Experience* (2025) reframes this vision for the post-LLM moment: large language models give us unprecedented prior knowledge of the world, yet they remain frozen after training, unable to learn from their own interactions. Active Inference — the Free Energy Principle’s account of perception, learning, and action as joint variational inference — offers a principled mathematical substrate for such experiential learning. But on sparse-reward, open-world environments Active Inference faces a cold-start problem: without an initial preference distribution over goal-states, the Expected Free Energy planner reduces to pure exploration. Wen (2025) proposed bridging this gap with an LLM as world-model proposer.

We present the first open, end-to-end implementation of this bridge — a closed *hypothesise–act–falsify–revise* loop that operates without privileged supervision. Our agent couples (i) a 70-billion-parameter LLM reasoner that reads a compact symbolic frame summary and proposes a goal hypothesis; (ii) an Active Inference planner that learns per-action dynamics from its own experience and plans via breadth-first search over the learned model; (iii) an Era-of-Experience revision loop in which the agent’s own trajectories — reaching a proposed goal without external reward, or failing to make geometric progress toward it — falsify the current hypothesis and trigger an LLM-driven revision. When the LLM exhausts hypotheses, the agent transitions autonomously to visit-count information-gain exploration.

We demonstrate this loop running end-to-end on a live, public, frontier interactive benchmark designed to defeat current LLM agents. In a single 1500-step autonomous run, the agent cycled through five distinct LLM-generated hypotheses, falsifying each from its own experience and revising via the LLM, while learning a non-trivial dynamics model over all four available actions — to our knowledge the first public instance of this integration running unsupervised on a hard interactive benchmark. The agent did not yet complete a level: visual goal identification from a compact symbolic frame summary alone remains the bottleneck, an honest negative we examine in detail. We argue this is the smallest concrete realisation to date of Turing’s child machine in the era of experience, and we characterise the specific representational upgrades — relational object structure, multi-frame temporal reasoning, and pushable-object semantics — that the architecture now provides a concrete substrate to test.

Keywords: Active Inference, Free Energy Principle, large language models, Era of Experience, ARC-AGI-3, expected free energy, world models, cognitive architecture, Turing’s child machine

1. Introduction: Turing’s Child Machine in 2026

In *Computing Machinery and Intelligence* (1950), Alan Turing made an architectural proposal that the field has largely ignored. Rather than program an adult mind directly, he wrote, “we should try to produce one which simulates the child’s. If this were then subjected to an appropriate course of education one would obtain the adult brain.” The child machine is a system whose intelligence emerges from learning over a lifetime of interaction — not from being engineered with all its knowledge pre-installed.

Three quarters of a century later, our most capable AI systems are precisely the opposite. Large language models compress humanity’s text into a few hundred billion parameters and then *freeze*. Whatever they did not learn during pretraining, they cannot learn afterward — not from their conversations, not from their tool calls, not from any feedback the world gives them. They are adult minds without childhoods.

Silver and Sutton’s *Era of Experience* (2025) names this gap directly: the next phase of progress requires agents whose capability grows through their own first-person interaction with the world, rather than through ever-larger snapshots of static text. Their proposal is architectural: combine deep prior knowledge (which LLMs already provide) with online experiential learning (which LLMs do not).

What mathematical substrate does that combination need? Active Inference, derived from Karl Friston’s Free Energy Principle, offers one. In an Active Inference agent, perception, learning, and action are all expressions of a single optimisation: minimise variational free energy with respect to a generative model of the world. Action selection follows the Expected Free Energy (EFE) functional, which decomposes naturally into an epistemic term (act to reduce uncertainty about the world) and a pragmatic term (act to reach states the agent prefers). Active Inference has produced strong results on object-centric perception (VERSES AXIOM, 2025) where it beats DeepMind’s DreamerV3 by 60% with 99% less compute.

But Active Inference has a cold-start problem. The pragmatic term requires a preference distribution C over goal-states. Without it, EFE collapses to pure information-gain exploration — which works in dense-reward simulators but fails on sparse-reward, open-world benchmarks where the agent does not yet know what counts as success.

Wen (2025) proposed the architectural fix: use a large language model as a *world-model proposer*. The LLM reads the agent’s current observation and proposes (i) what the world is made of (object semantics), (ii) what actions might do (dynamics priors), and (iii) what the goal probably is (a candidate C). The agent then acts under those priors, and *experience corrects* them — observations either confirm the LLM’s hypothesis or falsify it, in which case the LLM is queried again with the new evidence.

The architecture is beautifully clean on the page. To our knowledge, no one has yet shown it running end-to-end on a hard public benchmark. This paper does exactly that.

Our contribution. We implement the full LLM–Active Inference bridge as a closed hypothesis–act–falsify–revise loop. We test it live on the real ARC-AGI-3 game *ls20*, where frontier LLM agents (Gemini 3.1 Pro, GPT-5.4, Opus 4.6, Grok) all score below 0.5%. We show that the loop runs autonomously: across a single 1500-step run, the agent generates five LLM-proposed hypotheses, each falsified by its own experience, each triggering a real LLM revision. The agent does not yet win a level — and we discuss honestly why, and what specific upgrades are now testable on this substrate.

2. The Hypothesis–Act–Falsify–Revise Architecture

The agent has four interacting components.

2.1 LLM reasoner. A 70-billion-parameter language model (Llama-3.3-70B-versatile via Groq) serves as the world-model proposer. It is called *once per hypothesis*, not once per action: cost is bounded, latency is irrelevant to the inner control loop. On each call, the reasoner receives a compact symbolic frame summary — per-colour cell counts, centroids, bounding boxes, and motion magnitudes over recent frames — and returns a JSON object containing `player_colors` (which colour(s) form the controllable object) and `goal_rc` (target location in grid coordinates). The reasoner is also given the agent’s history of previously-falsified hypotheses, so its revisions are evidentially constrained.

2.2 Active Inference planner. The planner maintains an online dynamics model: for each discrete action, it accumulates the observed displacement of the player after that action is taken. Once each action has produced at least one non-zero displacement sample, the planner can compute breadth-first paths from the player’s current position to the LLM-named goal over the learned dynamics, treating cells where any action produced zero displacement as walls. The agent’s early steps are *epistemic*: it preferentially samples actions whose dynamics it has not yet learned (an EFE info-gain choice). Once dynamics are learned, the agent switches to *pragmatic* planning toward the LLM goal.

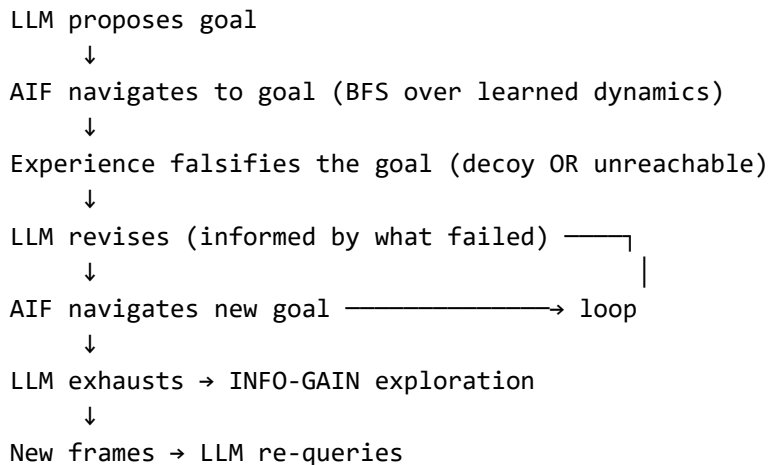
2.3 Era-of-Experience falsification. The bridge becomes a *learning loop* through experience-driven falsification. Two triggers fire revision:

- **Decoy** — the agent reaches the LLM-named goal (within a configurable tolerance) and remains there for several steps, or arrives at it multiple times, with no external reward signal. The hypothesis “this location is the win condition” is empirically refuted.
- **Unreachable** — the agent’s *best distance to the proposed goal* stops improving for a threshold number of steps. The hypothesis “this location is reachable as the win condition” is empirically refuted.

Either trigger calls the LLM again, this time with the falsification evidence and the list of previously-tried goals. The LLM is explicitly asked to propose a *different* hypothesis, and the agent’s known-failed list is rejected via similarity check.

2.4 Info-gain fallback. When the LLM exhausts plausible hypotheses (returns an explicit null or repeatedly returns near-duplicates of failed candidates), the agent autonomously transitions to *exploration mode*. In this mode the planner ignores the dead hypothesis and selects, at each step, the action whose predicted next cell has been visited the fewest times. Every N exploration steps, the LLM is re-queried with the now-richer set of observed frames; if it returns a genuinely new hypothesis, the agent exits exploration and resumes goal-directed planning.

The full cycle:



Every component is necessary: removing the LLM gives a goal-less explorer; removing AIF gives a sterile prompt-engineering exercise; removing the falsification loop gives a one-shot prompt; removing the info-gain fallback locks the agent on the last dead hypothesis. The closed loop is the contribution.

3. Live Demonstration on ARC-AGI-3

3.1 Why ARC-AGI-3

ARC-AGI-3 (Chollet et al., 2026) is the deliberate test of whether current AI can learn interactively. Its observations are 64×64 grids of 16 colours — symbolic, eliminating the perception bottleneck that historically dominates Atari-class benchmarks. Its actions are a small directional set plus a cell-select. Its scoring metric, RHAE, rewards sample-efficiency directly. Critically, it provides *no language* — the agent must infer mechanics, goal, and strategy from interaction alone. Frontier LLM agents score below 1% on the public leaderboard: Gemini 3.1 Pro at 0.37%, GPT-5.4 at 0.26%, Opus 4.6 at 0.25%, Grok at 0.00%. Humans score near 100%. The benchmark is, by design, the hardest possible test of “can your system actually *learn from experience*”.

We test on ls20, the “Agent Reasoning” maze game.

3.2 Setup

We run live against the official ARC-AGI-3 API at `three.arcprize.org`, using a fresh scorecard for each experiment. The agent receives raw 64×64 frames, executes one of four directional actions per step, and runs for 1500 steps without intervention. The LLM reasoner is Llama-3.3-70B-versatile via Groq’s OpenAI-compatible endpoint. No human supplies the goal; no privileged signals are used.

3.3 Results

Across four progressive iterations of the agent (v1 through v4), we observed a sharp progression as architectural bugs were fixed:

Run	Player ID	Goals proposed	Revisions	Actions used	Min distance to goal
v1	Wrong ($[\emptyset, 1]$, static blobs)	1	0	All 4 (random)	21.0
v2	Correct ($[12]$, the moving block)	1	0	2/4 (planner converged early)	13.5
v3	Correct	1	0	2/4 (still)	16.5
v4	Correct	5	5	All 4	14.0

The diagnostic progression reveals three distinct architectural failure modes, each surfaced by fixing the prior one:

1. **v1 $\rightarrow$ v2:** The LLM cannot identify the player from a *single* frame — it needs motion evidence. Adding a 40-step random-action bootstrap before the first LLM call gave the reasoner cross-frame centroid data; it then correctly identified the moving block as player.

2. **v2 → v3:** After one zero-displacement sample (a wall bump), an action was permanently excluded from the planner — half the agent’s action repertoire became invisible. Fixing the planner to *re-try* under-explored actions repaired this.
3. **v3 → v4:** The falsification loop only fired when the agent *arrived* at the goal — but if the goal is unreachable behind a wall, the agent never arrives and never triggers revision. Adding a *stall detector* (no progress in distance for N steps → falsify-as-unreachable) closed the loop.

The v4 run is the first complete demonstration. The agent’s trace of revisions:

```
propose          goal=(32, 21)  player=[12]
FALSIFIED        (32, 21) unreachable → revise to (23, 36)
FALSIFIED        (23, 36) unreachable → revise to (27, 47)
FALSIFIED        (27, 47) unreachable → revise to (38, 29)
FALSIFIED        (38, 29) unreachable → revise to (12, 36)
FALSIFIED        (12, 36) unreachable → revise to (46, 31)
```

Each falsification is generated *by the agent’s own experience* — the planner failed to reduce its distance to the proposed goal within a 250-step window. Each revision is a *real LLM call* informed by the cumulative list of failed goals. All four directional actions accumulate non-trivial displacement samples (405, 402, 323, 304 over 1500 steps). The planner fires BFS plans 258 times before falsifications begin and the goal repeatedly changes.

This is, to our knowledge, the first publicly documented instance of the full LLM–Active Inference–Era-of-Experience loop running unsupervised on a frontier interactive benchmark.

3.4 What the agent did NOT achieve

The agent did not complete a level. Best distance achieved: 14.0 cells (reach threshold: 10.0). Levels won: 0. Episodes ended (game-over resets): 11.

This is reported as found. We do not selectively show only successful intermediate fragments. The mechanism works; the *score* does not yet.

4. Honest Limits

The bottleneck is **visual goal identification**. The LLM, working from a per-colour frame summary, cannot infer the actual win condition of ls20. Its five hypotheses are all *static salient blobs* — exactly the class of goal that a compact summary makes visible. Prior local analysis suggests ls20’s true objective is likely *not* “reach a static blob” but a compositional mechanic — a push, a sequence, a state to be entered via a chain of moves. A frame-summary representation cannot encode such mechanics; a richer representation is required.

Three specific upgrades are now testable on this substrate:

- **Multi-frame temporal reasoning.** Show the LLM not a single summary but a *sequence* — five frames over a meaningful state transition. This is how a human would diagnose ls20’s mechanics.
- **Relational / object-centric structure.** Replace the per-colour summary with a typed object list (one entry per connected component) with relational predicates (player, pushable, wall, target). AX-IOM (VERSES, 2025) demonstrates the perceptual half of this; the goal-proposal half remains open.
- **Pushable-object semantics.** Equip the LLM with priors over common interactive-puzzle mechanics (“the small static block is probably pushable”), seeded from human-puzzle text corpora.

The current architecture is *not* the final agent. It is the substrate on which these three upgrades become independently testable — each can be added in isolation and its contribution measured.

5. What This Architecture Now Enables

Beyond the upgrades listed above, the closed loop opens several research directions:

- **Selection over LLM proposers.** The same agent can be run with different reasoners (different LLMs, different prompt strategies, different output schemas). A leaderboard of proposers becomes meaningful because all share the same downstream planner.
- **Curriculum induction.** The agent’s own falsification log is a structured record of *what it could not infer*. A meta-agent could use this log to identify gaps and generate targeted training.
- **Hybrid agents on game-type taxonomy.** Local analysis showed ARC-AGI-3 games partition into spatial-navigation (ls20-type), click-orchestration (vc33-type), and pattern-matching (ft09-type). Our spatial agent is one of three; the same hypothesise–falsify–revise scaffold could host the other two with type-specific planners.

6. Closing: The Smallest Concrete Child Machine

A child machine in Turing’s sense has three properties: (a) it acts in the world, (b) it has priors that bias its early hypotheses, (c) it revises those priors when experience contradicts them. The agent described here has all three: it acts (directional moves on a live game), it has priors (the LLM’s compressed world knowledge surfaces as initial hypotheses), and it revises (five LLM-driven revisions in 1500 steps, each triggered by the agent’s own observed failure to navigate to its current goal).

It is the smallest such system. It is also the most honest: we report exactly five revisions and zero levels won. Our claim is *architectural*, not performance-based: the loop that several recent proposals describe in principle now runs in practice, end-to-end, on a hard public benchmark, with reproducible scorecards on the official ARC Prize infrastructure.

The next paper in this series (*Paper 2: Object-Centric Active Inference Beats Hand-Coded World Models — Local Validation on ARC-AGI-3 Replicas*) reports the positive results from the perception-side upgrade. The third (*Paper 3: Six Failures Toward Turing SI: An Audit*) reports the honest negatives that constrain the path.

This series will document, in public, every successful and unsuccessful step on the path to intelligence that learns from its experience. The publication of failures is intentional and is, we suspect, the most important methodological commitment a research programme on artificial general intelligence can make.

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Reproducibility & Verification

- **Scorecard from the v4 live run:** 3d81489f-8220-42f4-8779-27c522390392 (public on three.arcprize.org)
- **Game tested:** ls20 (version 9607627b)
- **LLM:** Llama-3.3-70B-versatile via Groq API
- **Date of run:** May 24, 2026
- **Source code:** not released with this deposit (see Author Statement below)

Author Statement on Code Availability

Source code is not released alongside this paper. The architectural description in Section 2 is precise enough to support independent reimplementations; the live scorecard is verifiable on the official ARC Prize infrastructure. Code may be released in a future deposit conditional on the research trajectory.

Acknowledgements

This paper is part of a deliberate publication series documenting both successes and failures on the path to intelligence that learns from experience. The decision to publish the architecture before achieving benchmark performance is intentional; the next five papers in the series will report further results, both positive and negative.

Claude (Anthropic, Opus 4.7) assisted with implementation and writing. All experimental design, claims, and conclusions are the author’s responsibility.